

Advanced (Habanero)

- Q1.** Tom has \$15 to spend on tickets to a concert. The cost of one adult ticket is \$5 and one child ticket is \$3. If Tom buys x adult tickets and y child tickets, which system represents the total cost?
- (A) $5x + 3y = 15$
(B) $5x - 3y = 15$
(C) $3x + 5y = 15$
(D) $3x - 5y = 15$
(E) $5x + 5y = 15$
- Q2.** A bakery sells a total of 250 loaves of bread per day. They sell a combination of whole wheat and white bread. If they sell x whole wheat loaves and y white bread loaves, which system represents the total number of loaves sold?
- (A) $x + y = 250$
(B) $x - y = 250$
(C) $y - x = 250$
(D) $x + y = 500$
(E) $x - y = 500$
- Q3.** A car rental company charges a base fee of \$20 plus an additional \$0.25 per mile. If x is the number of miles driven and y is the total cost, which system represents the relationship between the miles driven and the total cost?
- (A) $y = 20 + 0.25x$
(B) $y = 20 - 0.25x$
(C) $y = 0.25x - 20$
(D) $y = 0.25x + 10$
(E) $y = 20 + 0.5x$
- Q4.** A person has \$1200 to invest in two types of accounts: one that earns 5% interest and another that earns 3% interest. If x is the amount invested in the 5% account and y is the amount invested in the 3% account, which system represents the total amount invested?
- (A) $x + y = 1200$
(B) $x - y = 1200$
(C) $y - x = 1200$
(D) $0.05x + 0.03y = 1200$
(E) $0.05x - 0.03y = 1200$
- Q5.** A company produces two products, A and B. Each product requires 2 hours of labor and 3 hours of machinery. If x is the number of units of product A and y is the number of units of product B, which system represents the total labor hours used?
- (A) $2x + 2y = 12$
(B) $2x + 3y = 12$
(C) $3x + 2y = 12$
(D) $2x + 2y = 6$
(E) $3x + 3y = 12$
- Q6.** A school is planning a field trip. The cost of the trip is \$25 per student and \$50 per chaperone. If x is the number of students and y is the number of chaperones, which system represents the total cost?
- (A) $25x + 50y = 1000$
(B) $25x - 50y = 1000$
(C) $50x + 25y = 1000$
(D) $25x + 25y = 1000$
(E) $50x - 25y = 1000$
- Q7.** A bookshelf has 5 shelves, and each shelf can hold 8 books. If x is the number of books on the shelf and y is the number of empty shelves, which system represents the relationship between the number of books and the number of empty shelves?
- (A) $x + 8y = 40$
(B) $x - 8y = 40$
(C) $8x + y = 40$
(D) $x + y = 40$
(E) $8x - y = 40$
- Q8.** A person has \$100 to spend on movie tickets and snacks. The cost of one movie ticket is \$10 and one snack is \$5. If x is the number of movie tickets and y is the number of snacks, which system represents the total cost?
- (A) $10x + 5y = 100$
(B) $10x - 5y = 100$
(C) $5x + 10y = 100$
(D) $10x + 10y = 100$
(E) $5x - 10y = 100$

Open Ended Questions (FRQ)

- Q9.** A company produces two products, X and Y. The production cost for X is \$10 per unit and for Y is \$15 per unit. The company has a budget of \$3000 for production. If x is the number of units of product X and y is the number of units of product Y, write a system of equations to represent the production costs and budget constraint.
- Q10.** A person has \$500 to invest in a savings account and a stock. The savings account earns 2% interest and the stock earns 5% interest. If x is the amount invested in the savings account and y is the amount invested in the stock, write a system of equations to represent the total investment and the interest earned.

Chef's Tips

- Encourage students to read the questions carefully and identify the key elements of the problem.
- Remind students to use mathematical models to represent real-world situations.
- Emphasize the importance of interpreting solutions in context and evaluating reasonableness.